

Assignment Management System — Backend Documentation

1. Overview

The Assignment Management System is a Spring Boot backend supporting two user roles — **Teacher** and **Student** — built around batches, assignments, and submissions. Teachers create batches and assignments, review submissions, and give feedback. Students self-register into an existing batch and submit assignment work.

Each feature below is shown as a **file-by-file flow diagram** — one box per file, in the exact order a request passes through them, with the file's full path and its job in that step.

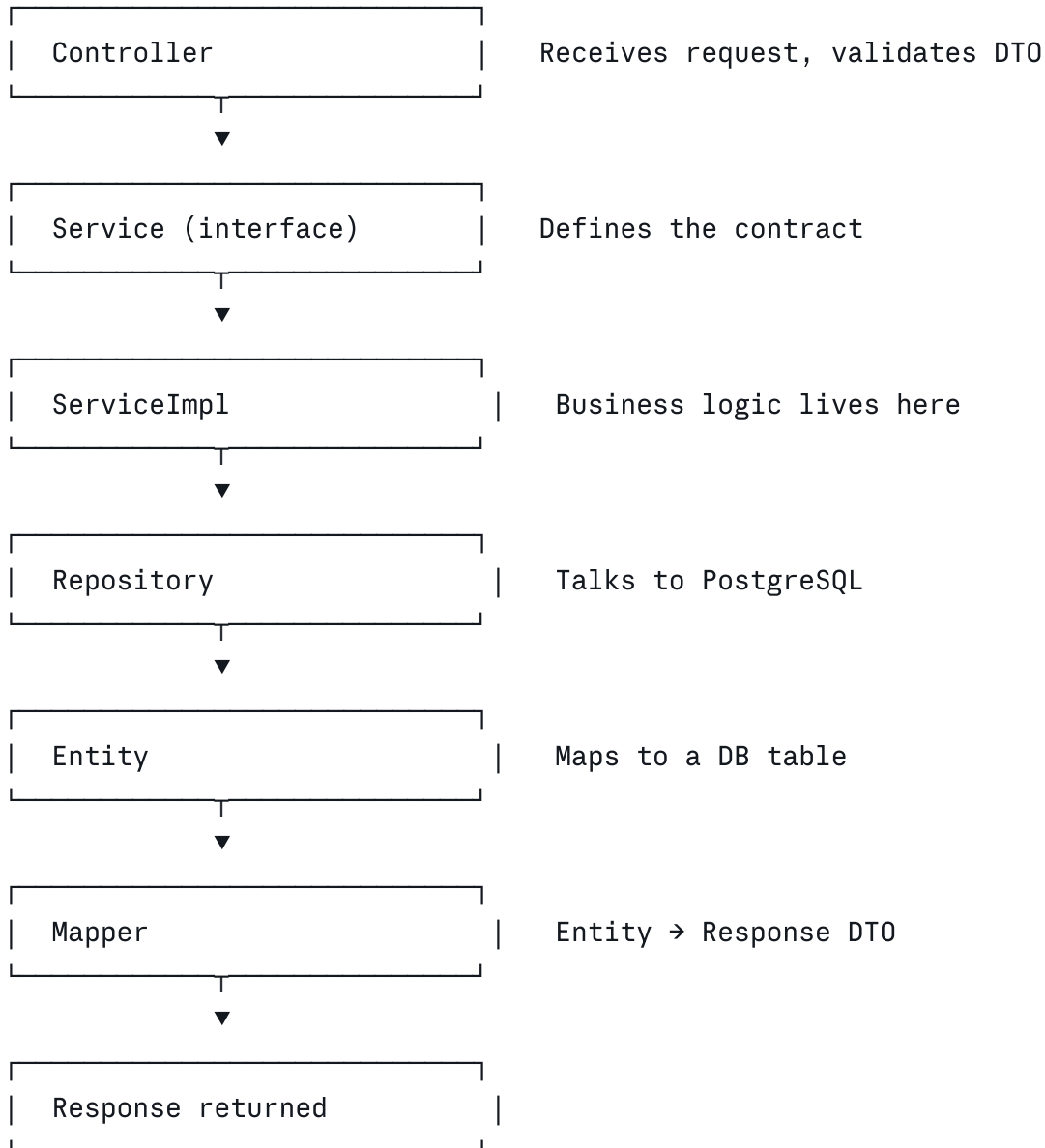
Base package for all paths below: `src/main/java/com/assignment/`

2. Architecture Layers

Layer	Responsibility	Path
Controller	Receives HTTP requests, validates input DTOs, delegates to services	<code>src/main/java/com/assignment/controller/</code>
Service (Interface)	Defines the business contract	<code>src/main/java/com/assignment/service/</code>
Service Implementation	Contains actual business logic	<code>src/main/java/com/assignment/service/impl/</code>
Repository	Data access via Spring Data JPA	<code>src/main/java/com/assignment/repository/</code>
Entity	Maps Java objects to PostgreSQL tables	<code>src/main/java/com/assignment/entity/</code>
Mapper	Converts between Entities and DTOs	<code>src/main/java/com/assignment/mapper/</code>
Request DTO	Defines API input contracts	<code>src/main/java/com/assignment/dto/request/</code>
Response DTO	Defines API output contracts	<code>src/main/java/com/assignment/dto/response/</code>

Layer	Responsibility	Path
Security	JWT-based authentication and user context	src/main/java/com/assignment/security/
Config	Security, Redis, Cloudinary, OpenAPI setup	src/main/java/com/assignment/config/
Exception Handling	Centralized error responses	src/main/java/com/assignment/exception/
Utility	Shared helper logic (JWT, file validation, Cloudinary)	src/main/java/com/assignment/util/
Enums	Fixed value sets (roles, statuses)	src/main/java/com/assignment/enums/
Constants	App-wide constant values	src/main/java/com/assignment/constants/

General request pipeline (every feature follows this shape):



3. Roles

Teacher

- Registers and logs in
- Creates batches
- Views batches they created
- Creates assignments for a specific batch
- Uploads assignment resources (PDF, PPT, images, ZIP, videos) via Cloudinary
- Views all submissions for an assignment
- Reviews submissions, assigns marks and feedback
- Sees which students have and haven't submitted

Student

- Self-registers (no teacher action required)
 - Selects an existing, active Batch ID during registration
 - Automatically linked to that batch upon successful registration
 - Submits assignments
-

4. Feature Flows

4.1 Teacher Registration

Endpoint: `POST /api/auth/register/teacher`

```
| src/main/java/com/assignment/controller/  
| AuthController.java  
| Receives request, runs validation
```



```
| src/main/java/com/assignment/service/impl/  
| AuthServiceImpl.java  
| Checks email, encrypts password
```



```
| src/main/java/com/assignment/mapper/  
| TeacherMapper.java  
| Converts request DTO to entity
```



```
| src/main/java/com/assignment/repository/  
| TeacherRepository.java  
| Persists teacher to PostgreSQL
```



```
| src/main/java/com/assignment/entity/  
| Teacher.java  
| Maps to the teachers table
```



```
| src/main/java/com/assignment/controller/  
| AuthController.java  
| Returns JWT + teacher details (AuthResponse)
```

Supporting components used in this flow:

File	Path
JwtService	src/main/java/com/assignment/config/JwtService.java
JwtUtil	src/main/java/com/assignment/util/JwtUtil.java
GlobalExceptionHandler	src/main/java/com/assignment/exception/GlobalExceptionHandler.java
BadRequestException	src/main/java/com/assignment/exception/BadRequestException.java

4.2 Batch Creation (Teacher)

Endpoint: POST /api/teacher/batch

```
| src/main/java/com/assignment/controller/ |  
| BatchController.java |  
| Receives request, resolves teacher from JWT |
```



```
| src/main/java/com/assignment/service/impl/ |  
| BatchServiceImpl.java |  
| Validates name, sets status ACTIVE |
```



```
| src/main/java/com/assignment/mapper/ |  
| BatchMapper.java |  
| Converts request DTO to entity |
```



```
| src/main/java/com/assignment/repository/ |  
| BatchRepository.java |  
| Persists batch with teacher reference |
```



```
| src/main/java/com/assignment/entity/ |  
| Batch.java |  
| Maps to the batches table |
```



```
| src/main/java/com/assignment/controller/ |  
| BatchController.java |  
| Returns BatchResponse |
```

Once a batch is **ACTIVE**, it becomes selectable during student registration — this is the connecting point between sections 4.2 and 4.3.

4.3 Student Self-Registration

Endpoint: **POST** /api/auth/register/student

Request Body

```
json
{
  "fullName": "Rahul Sharma",
  "email": "rahul@example.com",
  "password": "Password@123",
  "phone": "9876543210",
  "batchId": 5
}
```

Flow


```
| src/main/java/com/assignment/controller/ |  
| AuthController.java |  
| Receives request, runs field validation |
```



```
| src/main/java/com/assignment/service/impl/ |  
| AuthServiceImpl.java |  
| Verifies email uniqueness, validates batch ID |
```



```
| src/main/java/com/assignment/repository/ |  
| BatchRepository.java |  
| Confirms batch exists and is active |  
|  
| — if invalid —> BadRequestException.java |  
| ("Invalid Batch ID.", HTTP 400) |
```



```
| src/main/java/com/assignment/mapper/ |  
| StudentMapper.java |  
| Converts request DTO to entity (BCrypt password, |  
| batch reference attached) |
```



```
| src/main/java/com/assignment/repository/ |  
| StudentRepository.java |  
| Persists student to PostgreSQL |
```



```
| src/main/java/com/assignment/entity/ |  
| Student.java |  
| students table - @ManyToOne link to Batch.java |
```



```
| src/main/java/com/assignment/controller/ |  
| AuthController.java |
```

| Returns success response |

Success Response

```
json
{
  "message": "Student registered successfully.",
  "studentId": 12,
  "batchId": 5
}
```

Error Response — Invalid Batch

```
json
{
  "status": 400,
  "message": "Invalid Batch ID."
}
```

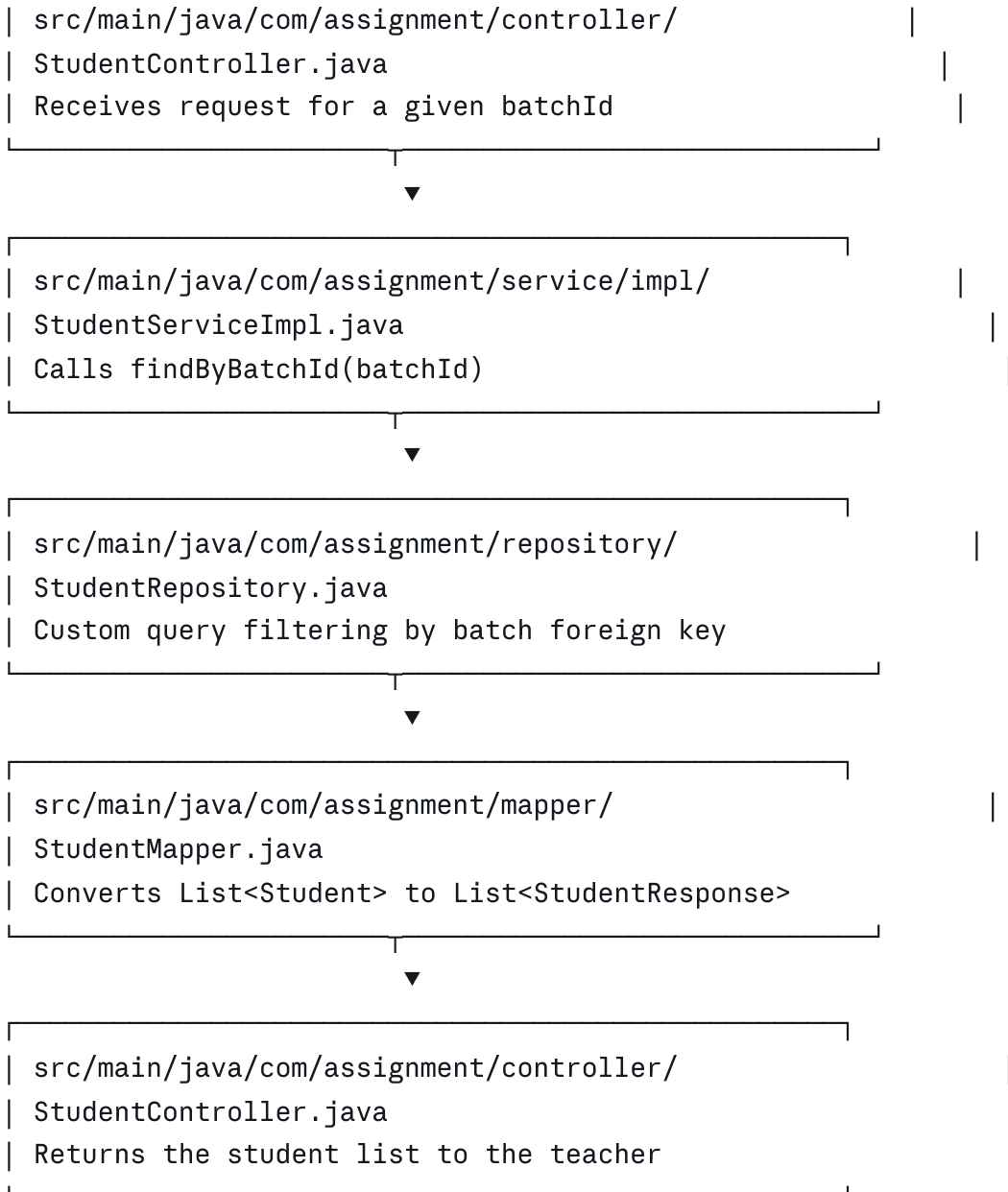
Backend Validation Rules

- Full name is required
- Email must be unique
- Password must satisfy the password policy
- Phone number is required
- Batch ID is mandatory and must reference an existing, active batch

Key architectural point: The `@ManyToOne` relationship lives on `entity/Student.java`, pointing to `entity/Batch.java`. This single relationship is what makes it possible for a teacher to immediately see which students belong to a batch — there is no separate "assign student to batch" step.

4.4 Teacher Viewing Students in a Batch

Endpoint: `GET /api/teacher/batch/{batchId}/students`



This is a read-only query against the foreign key set during student registration — no manual linking step exists.

4.5 Assignment Creation + Resource Upload (Teacher)

Endpoint: `POST /api/teacher/assignment`

Request Body (multipart/form-data)

json

```
{  
  "title": "Data Structures - Assignment 1",  
  "description": "Implement a singly linked list with insert/delete operations.",  
  "batchId": 5,  
  "assignmentType": "PDF",  
  "dueDate": "2026-07-20"  
}
```

(with an attached file field for the resource itself)

Flow

```
| src/main/java/com/assignment/controller/
| AssignmentController.java
| Receives request and uploaded file
```



```
| src/main/java/com/assignment/util/
| FileValidator.java
| Validates file type and size against AssignmentType.java
|
| — if invalid —> FileUploadException.java
```



```
| src/main/java/com/assignment/service/impl/
| AssignmentServiceImpl.java
| Validates the batch exists
```



```
| src/main/java/com/assignment/service/impl/
| CloudinaryServiceImpl.java
|
| Uploads file (via CloudinaryConfig + CloudinaryUtil),
|
| returns hosted file URL
```



```
| src/main/java/com/assignment/mapper/
|
| AssignmentMapper.java
|
| Converts request + Cloudinary URL to entity
```



```
| src/main/java/com/assignment/repository/
|
| AssignmentRepository.java
```

```
| Persists assignment to PostgreSQL
```

```
|
```

```
|-----|
```

```
|      ▼
```

```
|
```

```
| src/main/java/com/assignment/entity/
```

```
|
```

```
| Assignment.java
```

```
|
```

```
| Maps to assignments table (@ManyToOne to Batch & Teacher)
```

```
|
```

```
|-----|
```

```
|      ▼
```

```
|
```

```
| src/main/java/com/assignment/controller/
```

```
|
```

```
| AssignmentController.java
```

```
|
```

```
| Returns AssignmentResponse
```

```
|
```

```
|-----|
```

Supporting components:

File	Path
RedisService /	src/main/java/com/assignment/service/RedisService.java /
RedisServiceImpl	service/impl/RedisServiceImpl.java
RedisConfig	src/main/java/com/assignment/config/RedisConfig.java
GlobalExceptionHandler	src/main/java/com/assignment/exception/GlobalExceptionHandler.java

Endpoint: `POST /api/student/submission`

Request Body (multipart/form-data)

```
json
{
  "assignmentId": 21,
  "note": "Submitting my linked list implementation."
}
```

(with an attached file field for the submitted work)

Flow

| src/main/java/com/assignment/controller/
| SubmissionController.java
| Receives request and uploaded file



| src/main/java/com/assignment/util/
| FileValidator.java
| Validates the submitted file's type and size



| src/main/java/com/assignment/service/impl/
| SubmissionServiceImpl.java
| Checks assignment exists in student's batch,
| prevents duplicate submission



| src/main/java/com/assignment/service/impl/
| CloudinaryServiceImpl.java
|
| Uploads file, returns hosted URL



| src/main/java/com/assignment/mapper/
|
| SubmissionMapper.java
|
| Converts request + Cloudinary URL to entity



| src/main/java/com/assignment/repository/
|
| SubmissionRepository.java
|
| Persists submission to PostgreSQL



```
src/main/java/com/assignment/entity/
```

```
Submission.java
```

```
Maps to submissions table (@ManyToOne to Student & Assignment)
```



```
src/main/java/com/assignment/controller/
```

```
SubmissionController.java
```

```
Returns SubmissionResponse
```

4.7 Submission Review & Grading (Teacher)

Endpoint: `PUT /api/teacher/submission/{submissionId}/review`

Request Body

```
json
```

```
{  
  "marks": 8,  
  "feedback": "Good implementation, but missing edge case handling for empty li  
}
```

Flow

```
| src/main/java/com/assignment/controller/
| SubmissionController.java
| Receives marks and feedback
```



```
| src/main/java/com/assignment/service/impl/
| SubmissionServiceImpl.java
| Fetches submission, verifies teacher owns the assignment
|
| — if not found —> ResourceNotFoundException.java
```



```
| src/main/java/com/assignment/repository/
| SubmissionRepository.java
|
| Loads the existing submission
```



```
| src/main/java/com/assignment/mapper/
|
| SubmissionMapper.java
|
| Applies marks, feedback, updates status
|
| (SUBMITTED -> REVIEWED via SubmissionStatus.java)
```



```
| src/main/java/com/assignment/repository/
|
| SubmissionRepository.java
|
| Saves the updated submission
```



```
| src/main/java/com/assignment/entity/  
|  
| Submission.java  
|  
| submissions table row updated with marks/feedback/status  
|
```



```
| src/main/java/com/assignment/controller/  
|  
| SubmissionController.java  
|  
| Returns updated SubmissionResponse  
|
```

Related read endpoint — Submission Status Overview

GET /api/teacher/assignment/{assignmentId}/submissions

```
| src/main/java/com/assignment/controller/ |  
| SubmissionController.java                 |  
| Receives assignmentId                    |
```



```
| src/main/java/com/assignment/service/impl/ |  
| SubmissionServiceImpl.java                 |  
| Fetches all submissions for the assignment, |  
| cross-references with all students in the batch |  
| (via StudentRepository.java) to build submitted / |  
| not-submitted lists                        |
```



```
| src/main/java/com/assignment/mapper/      |  
| SubmissionMapper.java                     |  
| Converts both lists to SubmissionResponse objects |
```



```
| src/main/java/com/assignment/controller/  |  
| SubmissionController.java                 |  
|  
| Returns combined submitted/not-submitted view |
```

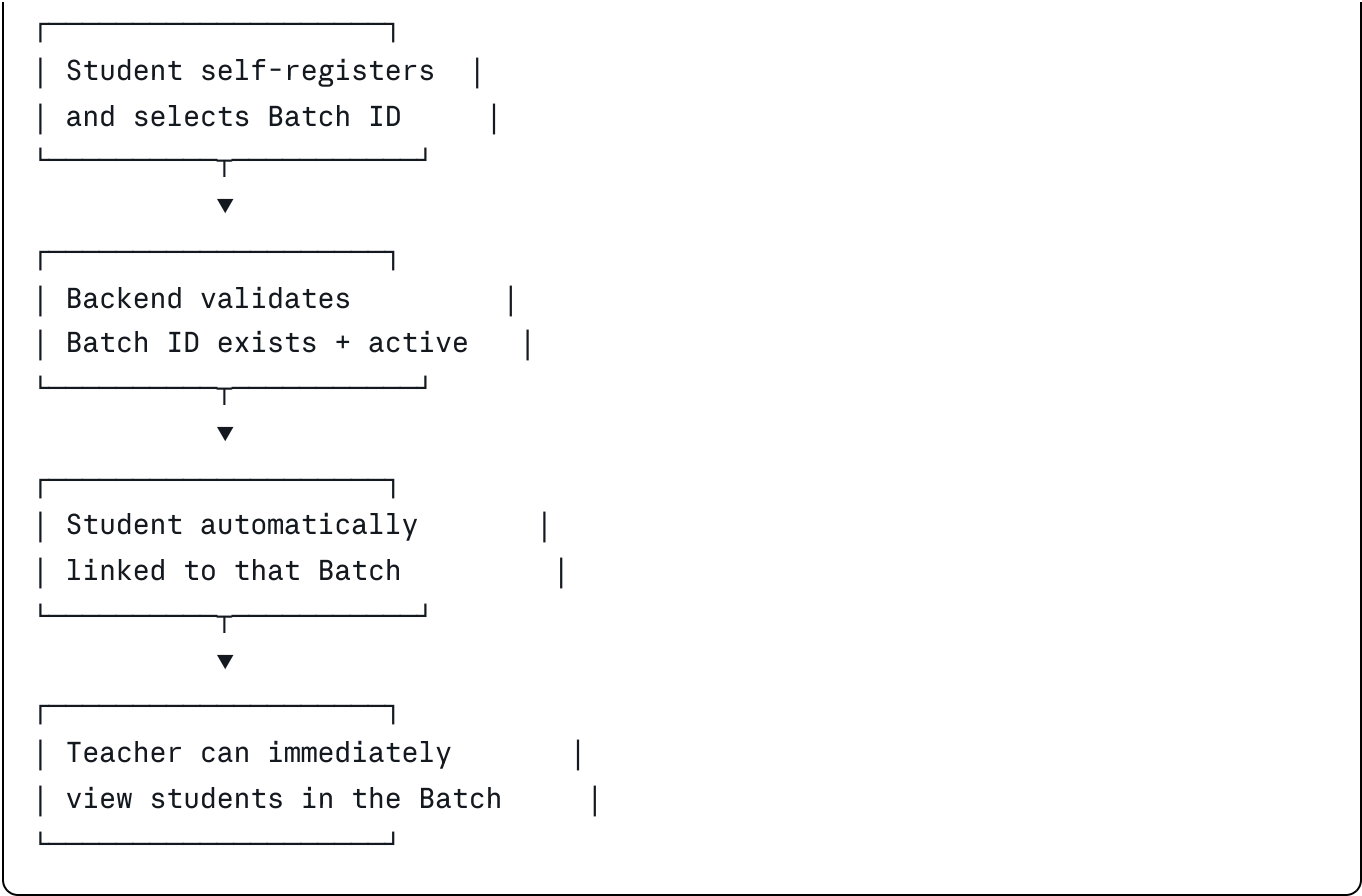
5. Batch-Student Relationship Summary

```
| Teacher creates |  
| Batch           |
```



```
| Batch becomes |  
| available (ACTIVE) |
```





The teacher never manually assigns students to a batch — the link is created entirely through student self-registration.

6. API Summary

Method	Endpoint	Role	Description
POST	<code>/api/auth/register/teacher</code>	Public	Teacher registration
POST	<code>/api/auth/register/student</code>	Public	Student self-registration into a batch
POST	<code>/api/auth/login</code>	Public	Login for both roles, returns JWT
POST	<code>/api/teacher/batch</code>	Teacher	Create a new batch
GET	<code>/api/teacher/batch</code>	Teacher	View all batches

Method	Endpoint	Role	Description
			created by this teacher
GET	<code>/api/teacher/batch/{batchId}/students</code>	Teacher	View students linked to a batch
POST	<code>/api/teacher/assignment</code>	Teacher	Create assignment + upload resource file
POST	<code>/api/student/submission</code>	Student	Submit assignment work
GET	<code>/api/teacher/assignment/{assignmentId}/submissions</code>	Teacher	View submitted / not-submitted students
PUT	<code>/api/teacher/submission/{submissionId}/review</code>	Teacher	Give marks and feedback on a submission

7. Cross-Cutting Components Reference

Component	Path
SecurityConfig	<code>src/main/java/com/assignment/config/SecurityConfig.java</code>
JwtAuthenticationFilter	<code>src/main/java/com/assignment/config/JwtAuthenticationFilter</code>

Component	Path
JwtAuthenticationEntryPoint	src/main/java/com/assignment/config/JwtAuthenticationEntry
JwtService	src/main/java/com/assignment/config/JwtService.java
RedisConfig	src/main/java/com/assignment/config/RedisConfig.java
CloudinaryConfig	src/main/java/com/assignment/config/CloudinaryConfig.java
OpenApiConfig	src/main/java/com/assignment/config/OpenApiConfig.java
GlobalExceptionHandler	src/main/java/com/assignment/exception/GlobalExceptionHanc
CustomUserDetails / CustomUserDetailsService	src/main/java/com/assignment/security/
AppConstants	src/main/java/com/assignment/constants/AppConstants.java

This document is intended as an onboarding reference. Each flow is shown as a box-per-file diagram in call order, with exact paths and no raw code, so it stays accurate as implementation details evolve.